Junseo Lee – *Curriculum Vitae*

Contact

David E. and Stacey L. Goel Quantum Science and Engineering Building (Harvard Quantum Initiative)
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Research Interests

Quantum Many-Body Physics, Quantum Information, Complexity Theory, Learning Theory, Hardness of Approximation

Education

Harvard University, Kenneth C. Griffin Graduate School of Arts and Sciences (GSAS) — Cambridge, MA
Doctor of Philosophy (Ph.D.) in Quantum Science and Engineering — Sep. 2026 – Present

- Advisors: Prof. Anurag Anshu and Prof. Sitan Chen (Computer Science); Prof. Jordan Cotler (Physics)
- Affiliations: Harvard Quantum Initiative; Theory of Computation Group (School of Engineering and Applied Sciences)
- Supported by the Kwanjeong Doctoral Study Abroad Fellowship

Massachusetts Institute of Technology (MIT) — Cambridge, MA
Cross-Registered Doctoral Student — Sep. 2026 – Present

Yonsei University — Seoul, Korea
Bachelor of Science (B.S.) in Electrical and Electronic Engineering — Mar. 2019 – Feb. 2023

- Thesis: *Combinatorial designs for information theory*
- Supported by the Hyundai Motor CMK Science and Technology Scholarship

Chungnam Science High School — Gongju, Korea
Concentration in Mathematics; *Early Honors Graduate* — Mar. 2017 – Dec. 2018

Research Positions

NSF AI Institute for Artificial Intelligence and Fundamental Interactions (IAIFI) — Cambridge, MA
Junior Investigator (Nominator/Advisor: Prof. Jordan Cotler) — May. 2026 – Present

Seoul National University (SNU) — Seoul, Korea
Research Fellow, Institute of Computer Technology (Host: Dr. Kabgyun Jeong) — Apr. 2026 – Aug. 2026
Research Affiliate, Research Institute of Mathematics (Host: Dr. Kabgyun Jeong) — Jan. 2023 – Mar. 2026
Undergraduate Researcher, Research Institute of Mathematics (Advisor: Dr. Kabgyun Jeong) — Mar. 2020 – Dec. 2022

Republic of Korea Army — Seoul, Korea
Research Scientist (Technical Research Personnel, Mandatory Military Service) — Mar. 2023 – Mar. 2026

Yonsei University — Seoul, Korea
Undergraduate Researcher, High Dimensional Signal Processing Lab (Advisor: Prof. Chulhee Lee) — Jul. 2022 – Dec. 2022
Undergraduate Researcher, Mathematical Biology Lab (Advisor: Prof. Jeehyun Lee) — Jan. 2022 – Jun. 2022

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, Korea
Pre-Undergraduate Researcher, Atomic-Scale Device Simulation Lab (Advisor: Prof. Yong-Hoon Kim) — Summer 2018
Pre-Undergraduate Researcher, Therapeutic Biomaterials Engineering Lab (Advisor: Prof. Ji-Ho Park) — Summer 2017

Publications (Google Scholar Profile)

*Equal contribution. †Authors listed alphabetically.

(16) M. Shin, **J. Lee**, and C. Oh, "Heisenberg-limited Hamiltonian learning without short-time control," (2026). arXiv:2604.27838.
   - Long talk at the *26th Asian Quantum Information Science Conference* (**AQIS 2026**).

(15) A. Bluhm, M. C. Caro, F. E. Gutiérrez, **J. Lee**†, A. Oufkir, C. Rouzé, and M. Shin, "Certifying and learning local quantum Hamiltonians," (2026). arXiv:2603.29809.
   - Extends and subsumes my earlier work arXiv:2512.09778.
   - Contributed talk at the *21st Conference on the Theory of Quantum Computation, Communication and Cryptography* (**TQC 2026**).

(14) **J. Lee**† and M. Shin, "Optimal certification of constant-local Hamiltonians," (2025). arXiv:2512.09778.

(13) M. Fanizza, V. Iyer, **J. Lee**†, A. A. Mele, and F. A. Mele, "Efficient learning of bosonic Gaussian unitaries," (2025). arXiv:2510.05531.
   - Contributed talk at the *29th Annual Conference on Quantum Information Processing* (**QIP 2026**).

(12) D. Ji, **J. Lee**, M. Shin, I. Sohn, and K. Jeong, "Bounding quantum uncommon information with quantum neural estimators," *Quantum Science and Technology* **11**, 015001 (2026). DOI: 10.1088/2058-9565/ae18f4.

(11) K. Anand, K. Jeong, and **J. Lee**†, "Collapses in quantum-classical probabilistically checkable proofs and the quantum polynomial hierarchy," (2025). arXiv:2506.19792.

(10) M. Shin*, **J. Lee***, S. Lee, and K. Jeong, "Resource-efficient algorithm for estimating the trace of quantum state powers," *Quantum* **9**, 1832 (2025). DOI: 10.22331/q-2025-08-27-1832.

(9) M. Lee, M. Shin, **J. Lee**, and K. Jeong, "Mutual information maximizing quantum generative adversarial networks," *Scientific Reports* **15**, 32835 (2025). DOI: 10.1038/s41598-025-18476-y.

(8) M. Shin*, S. Lee*, **J. Lee***, D. Ji, H. Yeo, and K. Jeong, "Disentanglement provides a unified estimation for quantum entropies and distance measures," *Physical Review A* **110**, 062418 (2024). DOI: 10.1103/PhysRevA.110.062418.

(7) M. Shin, **J. Lee**, and K. Jeong, "Estimating quantum mutual information through a quantum neural network," *Quantum Information Processing* **23**, 57 (2024). DOI: 10.1007/s11128-023-04253-1.

(6) **J. Lee** and K. Jeong, "Quantum Rényi entropy functionals for bosonic gaussian systems," *Physics Letters A* **490**, 129183 (2023). DOI: 10.1016/j.physleta.2023.129183

(5) **J. Lee**, H. Yeo, and K. Jeong, "Weighted p-Rényi entropy power inequality: Information theory to quantum Shannon theory," *International Journal of Theoretical Physics* **62**, 253 (2023). DOI: 10.1007/s10773-023-05512-8

(4) **J. Lee** and K. Jeong, "High-dimensional private quantum channels and regular polytopes," *Communications in Physics* **31**, 189 (2021). DOI: 10.15625/0868-3166/15762

(3) K. Jeong, **J. Lee**, *et al.*, "Single qubit private quantum channels and 3-dimensional regular polyhedra," *New Physics: Sae Mulli* **68**, 232 (2018). DOI: 10.3938/NPSM.68.232
   - Bronze Award, The Humantech Paper Award, Samsung Electronics (2018).

Book Chapters

(2) **J. Lee**, "Assessing Quantum Integer Factorization Performance with Shor's Algorithm,"
In: *Quantum Computing: A Journey into the Next Frontier of Information and Communication Security*,
Edited by Mohammad Hammoudeh, Abdullah T. Alessa, Amro M. Sherbeeni, Clinton M. Firth, Abdullah S. Alessa, *CRC Press* (2024). DOI: 10.1201/9781003475286

Patents

(1) K. Jeong, M. Shin, and **J. Lee**, "Method for estimating quantum mutual information through a quantum neural network," *Korea Patent* Open No. 10-2026-0009068 (2024). DOI: 10.8080/1020240091151

Professional Activities

Journal Reviewer

Physical Review Letters, PRX Quantum, Physical Review Research, Physical Review Applied, Physical Review A, npj Quantum Information, IEEE Transactions on Information Theory, Quantum, Physics Letters A, Annalen der Physik, Mathematical Methods in the Applied Sciences

Conference Reviewer

- *IEEE Symposium on Foundations of Computer Science* (FOCS), 2026
- *Theory of Quantum Computation, Communication and Cryptography* (TQC), 2026
- *Quantum Computing Theory in Practice* (QCTiP), 2026
- *Quantum Techniques in Machine Learning* (QTML), 2025

Organizing

- Organizer, *QISCA Winter School on Quantum Learning Theory for Bosonic and Fermionic Systems*, 2026
- Co-organizer, *SNU Quantum Information Theory Seminar*, 2024–2025
- Co-organizer, *Quantum AI Hackathon* (jointly organized by Kakao Enterprise and Jeonju University), 2025

Committee Service

- Poster Session Committee Member, *National Undergraduate Quantum Conference*, Seoul National University, 2026
- Selection Committee Member, *Quantum Internship Program* (National Information Society Agency; Korea Quantum Industry Center), 2024 – 2025

Outreach and Community Resources

- Creator and Maintainer, *Quantum Learning Theory Zoo*, 2025–present
- Facilitator (Mentor), *Korea Scholar's Conference for Youth (KSCY)*, Yonsei University, 2019

Selected Honors and Awards

Funding and Fellowships

- *Doctoral Study Abroad Fellowship*, Kwanjeong Educational Foundation, 2026–2030
- *Academic Travel Grant*, Hyundai Motor CMK Foundation, 2022
- *Hyundai Motor CMK Science and Technology Scholarship*, 2021–2022

Academic Excellence

- *High Honors*, Yonsei University, 2022
- *Honors*, Yonsei University, 2020–2021
- *Science Scholarship*, JH Foundation, 2017
- *Academic Excellence Scholarship*, Asan Future Scholarship Foundation, 2016

Additional Honors and Awards

- *Selected Paper Award*, Finance and Economics Contest (DB Group), 2022
- *Best Tutor Award*, Yonsei University, 2021–2022
- *Third Prize*, Undergraduate Research Exhibition, Korean Physical Society, 2021
- *Bronze Award*, Humantech Paper Award (Samsung Electronics), 2018
- *Best Translator Award*, NAVER Connect Foundation & Khan Academy, 2018
- *Pre-Undergraduate Research Fellowship*, Korea Advanced Institute of Science and Technology, 2017–2018
- *Honorable Mention (National) & Gold Award (Regional)*, Korean Olympiad in Informatics, 2016

Teaching

Special Lecture Series, Quantum Information Science Club Association

(Teaching materials available at: harris-junseo-lee.github.io/teaching/)

- Organizer and Lecturer, *Quantum Learning Theory for Bosonic and Fermionic Systems*, Lecture 6: "Learning Bosonic Gaussian Unitaries," Winter 2026
- Invited Lecturer, *Quantum Complexity Reading Group*, Fall 2025
- Invited Lecturer, *Quantum Learning and Complexity Theory*, Summer 2025

University–Industry Research Internship

- Instructor, *AAA559: College of Informatics Internship (II) (Graduate Course)*, Korea University, Fall 2025
- Instructor, *AAA558: College of Informatics Internship (I) (Graduate Course)*, Korea University, Fall 2025
- Instructor, *SW4343: Software Field Placement (I)*, Korea Aerospace University, Fall 2024

Yonsei University

- Teaching Assistant, *YCS1009: Change the World through Programming*, Fall 2022
- Teaching Assistant, *YCS1002: Software Programming*, Fall 2022
- Teaching Assistant, *EEE1108: Engineering Information Processing*, Fall 2021
- Course Tutor, *MAT2016: Engineering Mathematics (III)*, Spring 2022 (Best Tutor Award)
- Course Tutor, *MAT1012: Engineering Mathematics (II)*, Fall 2021 (Best Tutor Award)

Undergraduate Research Mentoring

- Hugo Mackay (Harvard College, 2026), Herchel Smith Undergraduate Science Research Program (in collaboration with Prof. Anurag Anshu)
- Arul Rhik Mazumder (Carnegie Mellon University, 2026)
- Jaemin Park (Seoul National University, 2026)
- Donghwa Ji (Seoul National University, 2025–2026, in collaboration with Dr. Kabgyun Jeong)
- Kartik Anand (Indian Institute of Technology Goa, 2025)
- Mingyu Lee (Seoul National University, 2024–2026), Summer Internship at the University of Oxford (2026, in collaboration with Dr. Sathyawageeswar Subramanian)
- Myeongjin Shin (KAIST, 2023–2026), Caltech Summer Undergraduate Research Fellowship (2026, in collaboration with Prof. John Preskill and Prof. Yu Tong)

Research Visits

- Harvard University (funded by Prof. Anurag Anshu and the Harvard Quantum Initiative), Mar. 2026

Selected Talks

*Online talk.

Research Talks

"Heisenberg-limited Hamiltonian learning without short-time control"

- Invited talk, Hamburg University of Technology (Group of Prof. Martin Kliesch), Jun. 2026*
- Long contributed talk, *26th Asian Quantum Information Science Conference* (**AQIS 2026**), Aug. 2026

"Certifying and learning local quantum Hamiltonians"

- Invited talk, Quantum Meets (invited by Prof. Atul Singh Arora), Jun. 2026*
- Invited talk, *7th QNRC Joint Workshop*, Korea Institute of Science and Technology Information, May 2026
- Contributed talk, *21st Conference on the Theory of Quantum Computation, Communication and Cryptography* (**TQC 2026**), Sep. 2026

"Optimal certification of constant-local Hamiltonians"

- Invited talk, *Quantum Software Lab Seminar*, University of Edinburgh, Mar. 2026*
- Contributed talk, *Annual Meeting of the Quantum Information Society of Korea* (QISK 2026), Feb. 2026

"Efficient learning of bosonic Gaussian unitaries"

- Invited talk, *Annual Meeting of the Quantum Information Society of Korea* (QISK 2026), Feb. 2026
- Invited talk, *N^3etFraST Workshop*, Nov. 2025
- Invited talk, *Quantum Data Science & AI Lab Seminar*, Yonsei University, Nov. 2025
- Contributed talk, *29th Annual Conference on Quantum Information Processing* (**QIP 2026**), Jan. 2026 (presented under the title "Efficient Learning Algorithms for Structured Bosonic and Fermionic Unitary Operators")

"Resource-efficient algorithm for estimating the trace of quantum state powers"

- Invited talk, Electronics & Telecommunications Research Institute (ETRI), Dec. 2024
- Invited talk, *Quantum Information Theory Seminar*, SNU, Dec. 2024*
- Invited talk, *IBM-Yonsei Qiskit Fall Fest*, Yonsei University, Nov. 2024*
- Contributed talk, *Annual Meeting of the Korean Mathematical Society* (KMS), Oct. 2024
- Poster, *28th Annual Conference on Quantum Information Processing* (QIP 2025), Feb. 2025

"Mutual information maximizing quantum generative adversarial network"

- Invited talk, *Triangle Quantum Computing Seminar*, NC State University Quantum Initiative, Nov. 2023*

"Estimating quantum mutual information through a quantum neural network"

- Invited talk, *CS Katha Barta*, National Institute of Science Education and Research Bhubaneswar, Aug. 2023*

"Quantum Rényi entropy functionals for bosonic Gaussian systems"

- Poster, *25th Annual Conference on Quantum Information Processing* (QIP 2022), Mar. 2022

"High-dimensional private quantum channels and regular polytopes"

- Invited talk, *KISTI-KU-SNU Joint Workshop*, Sep. 2023*
- Invited talk, *Quantum Information Theory Seminar*, SNU, Aug. 2021*
- Contributed talk, *Winter Meeting of the Optical Society of Korea* (OSK), Feb. 2022
- Contributed talk, *Fall Meeting of the Korean Physical Society* (KPS), Oct. 2021*
- Poster, *25th Annual Conference on Quantum Information Processing* (QIP 2022), Mar. 2022

Tutorials and Public Lectures

"Getting Started in Quantum Information Science"

- Invited talk, *Student Outreach Lecture*, Shinil High School, Apr. 2026
- Invited talk, *Student Outreach Lecture*, Shinil High School, Aug. 2024

"Learning theory in ∞-dimensional quantum systems"

- Invited talk, *Team QST Summer Workshop*, SNU, Aug. 2025

"Introduction to quantum machine learning"

- Invited talk, *Healthcare & Research Team Seminar*, Amazon Web Services (AWS) Korea, Mar. 2025

"Quantum machine learning models for drug library generation"

- Invited talk, *Quantum Computing and Monte Carlo Workshop*, Yonsei University, Aug. 2024

"$\mathsf{QMA} \stackrel{?}{=} \mathsf{NP}$: The NLTS theorem and the quantum PCP conjecture"

- Invited talk, *Center for Quantum Network's Channel Capacity Summer Workshop*, SNU, Jul. 2024

"Minimal data may be sufficient for quantum artificial intelligence"

- Invited talk, *Department of Mathematical Sciences Seminar*, SNU, Jun. 2023*